25901649 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCCFS102 - INTRODUCTION TO COMPUTER NETWORKS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **1** marks

1. Discuss the significance of the Application Layer in TCP/IP model. [U] / [CO1]
2. Identify the main purpose of network standards. [U] / [CO1]
3. Illustrate the role of the Physical Layer in data transmission [An] / [CO1]
4. Outline the device responsible for connecting different networks. [An] / [CO1]
5. Categorize the policies used in open-loop congestion control. [An] / [CO2]
6. Illustrate what happens with a protocol, when an acknowledgement packet is lost, within a noisy channel. [An] / [CO2]
7. Relate the term zone to its in DNS. [U] / [CO2]
8. Identify the full form of OSI [U] / [CO1]
9. Explain a passive attack [U] / [CO1]

10. Identify the purpose of an access control model [U] / [CO1]

11. Summarize why maintaining a proper chain of custody is important while handling digital evidence [U] / [CO4]

12. Discuss deadlock in an OS [U] / [CO4]

[1x10 = 10]

Part B

Short Essay Type Questions

Answer any **3** questions

Each question carries **5** marks

13. Explain why fiber optics are preferred in MANs. [U] / [CO1]

14. Explain SMTP. [U] / [CO2]

15. Compare and contrast VRC and LRC in error detection. [An] / [CO2]

16. Explain Security Mechanisms and list its types [U] / [CO1]

[5x3 = 15]

Part C

Essay Type Questions

Answer any **1** questions

Each question carries **10** marks

17. Illustrate protocol and standard and explain the key elements of protocol [An] / [CO1]

18. Explain the process of CRC error detection using suitable example. [U] / [CO2]

[10x1 = 10]